Data Visualization Assignment

DSA 301 | Project 2 Report

Japan Immigration Dataset & WHOOP Health Dataset

Mert Güngör 34159

https://github.com/mert-gng-99/DSA301-Project/tree/main/proj2

Part 1: Japan Inbound Immigration Dashboard

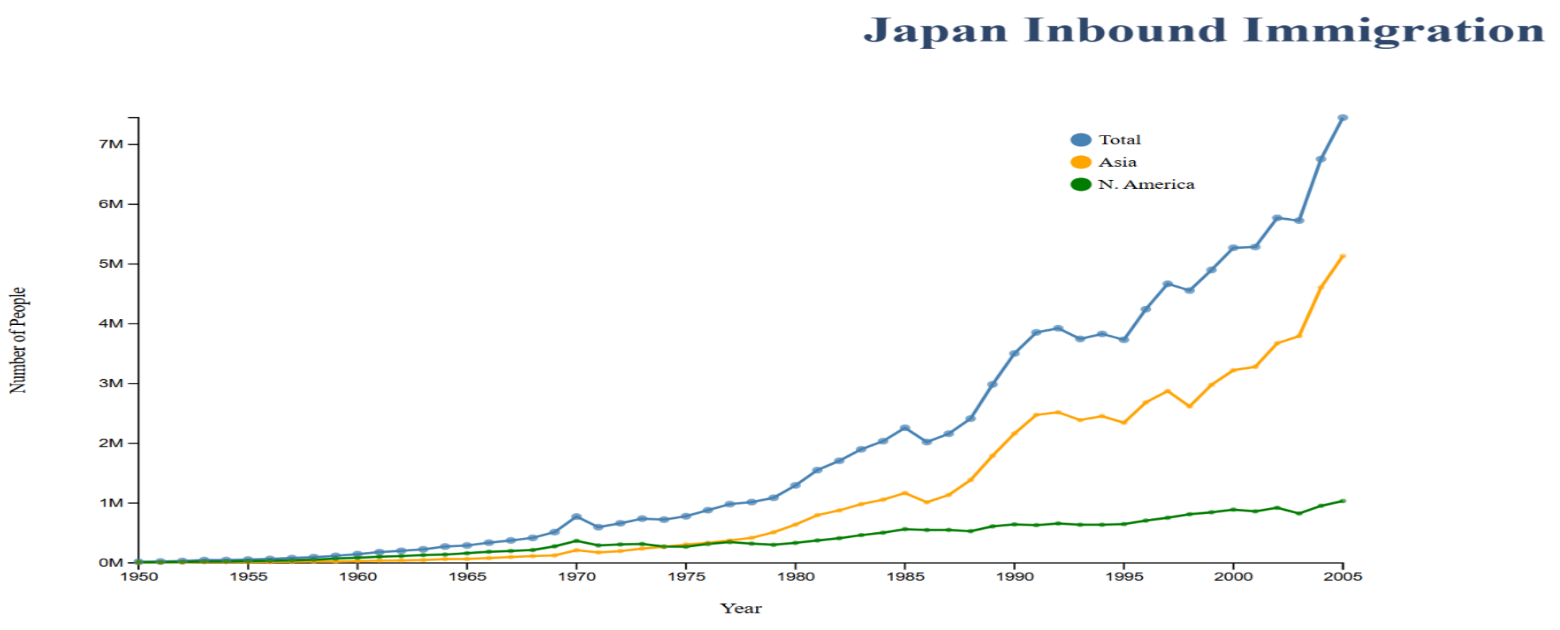
1A | First Dashboard and Prompt

I downloaded the Japan inbound immigration data from Kaggle. The dataset has information about how many people came to Japan from different countries every year, from 1950 to 2005.

I used this prompt to make the first dashboard:

```
"Create a line chart that shows total immigration to Japan over time from 1950
to 2005. Show three lines: total, Asia, and North America. Add colored circles
on the total line. Use steelblue, orange, and green for the lines."
```

The first dashboard shows a line chart with three lines. The X axis is year and the Y axis is total number of people. I also added a simple legend.



1B | Analysis: Gestalt and Bertin Principles

In this section I will analyze what is good and what can be improved in the first dashboard.

What is already good:

- Continuity (Gestalt): The line chart uses continuous lines. This helps the user to follow the trend over time. It is easy to see that immigration grew a lot after 1985.
- Position (Bertin): The Y axis uses position to show the number of people. This is the strongest visual variable, so quantitative data is encoded well.
- Hue (Bertin): Different colors are used for each region line. This makes it easy to separate Total, Asia, and North America from each other.
- Proximity (Gestalt): The legend is placed near the lines. This helps the user to understand which line is which.

What can be improved:

- Color consistency: The colors in the line chart are not consistent with a country bar chart. When the user sees a bar chart, the colors should mean the same thing as in the line chart (Bertin principle of consistent retinal variables).
- No tooltip: When you hover over a data point, nothing happens. The user cannot see the exact number. This is a problem because the Y axis labels are in millions and hard to read precisely.
- No year filter: There is no way to look at a specific year. The chart shows all years together which can be confusing.
- Figure ground (Gestalt): All three lines have similar visual weight. The "Total" line is the most important, but it does not stand out more than the others. It should be thicker or darker.
- Y axis tick format: The Y axis use M for scaling (like 1M, 2M, 7M) but it does not explain this means millions. A new user may not understand the scale at first look.

1C | Revised Dashboard

I made a new version of the dashboard to fix the problems I found. Here is the new prompt I used:

New prompt:

```
"Improve the Japan immigration dashboard. Add: 1) a year dropdown filter,
2) a second bar chart showing top 6 countries for the selected year,
3) tooltip on hover showing exact year and total, 4) consistent colors
across both charts, 5) thicker total line (2.5px), 6) proper axis labels,
7) sort bar chart from highest to lowest."
```

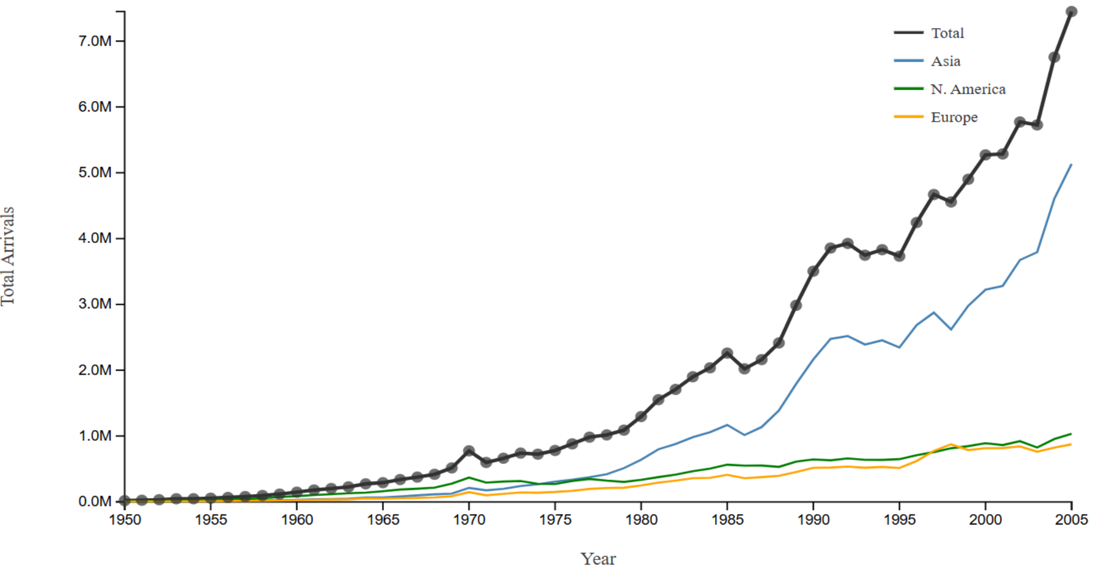
What is better now:

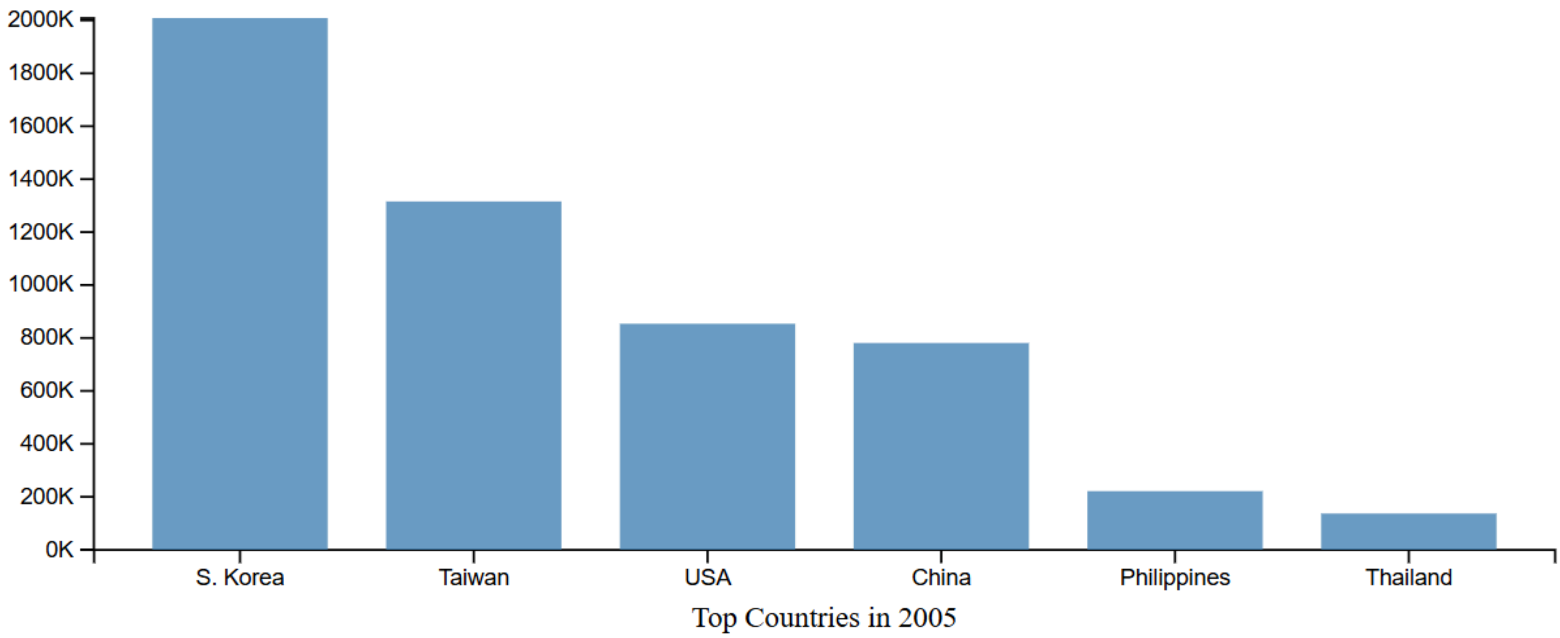
- I added a tooltip. Now when you hover over a data point, you can see the exact year and total number. This makes the chart more interactive.
- I added a year dropdown. The user can now select a year (1980, 1985, 1990, 1995, 2000, 2005) and see a bar chart for that year. This gives context.

- The total line is now thicker (2.5px) and darker. It stands out more than the region lines. This follows the figure ground principle.
- The bar chart sorts countries from highest to lowest. This makes it easy to compare countries (continuation and good data ink ratio).
- Colors are now consistent. Asia is always steelblue, North America is always green, Europe is always orange.

Japan Inbound Immigration - V2 (Improved)

Select Year: 2005





Part 2: WHOOP Health Dataset

2A | Dataset Analysis for a Mobile App

The WHOOP Health Dataset has 100,000 rows and 39 columns. It contains health and fitness data from many users. I will explain which attributes are useful for a mobile app and why.

Most useful attributes:

- recovery_score (0-100): This is the most important attribute. It tells the user how ready they are to exercise today. It should be shown very big on the home screen of the app. Users open the app every morning to see this number.
- hrv (Heart Rate Variability in ms): HRV is the main signal that the device uses to calculate recovery. Higher HRV means the body is more recovered. It is useful to show a 7-day trend chart of HRV so the user can see if they are getting better or worse.
- resting_heart_rate (bpm): If the resting heart rate is higher than usual, it can mean the user is sick or overtraining. It is good to show this next to HRV so the user can understand their recovery better.
- sleep_hours, sleep_efficiency, deep_sleep_hours, rem_sleep_hours: Sleep data is very important. The app should show a sleep breakdown bar so the user can see how much time they spent in each sleep stage.
- day_strain (0-21): This shows how hard the day was on the body. It is useful to show alongside recovery so the user can plan how hard to train.
- activity_type and activity_duration_min: These are useful for the workout history screen. The user can see what they did each day.
- hr_zone_1_min to hr_zone_5_min: Heart rate zone data is useful for athletes. It shows how many minutes were spent in each intensity zone during a workout.
- day_of_week: Useful for weekly overview and patterns. Some people always sleep worse on certain days.

Less useful for mobile display:

- user_id: This is only an ID, not useful to show.
- weight_kg and height_cm: These are static. They belong in the profile settings screen, not the main dashboard.
- workout_time_of_day: This is interesting but not critical for the main dashboard.

2B | Three Dashboard Sketches

I designed three different mobile dashboard sketches. Each one follows Gestalt and Bertin principles.

Sketch 1: Recovery Focus

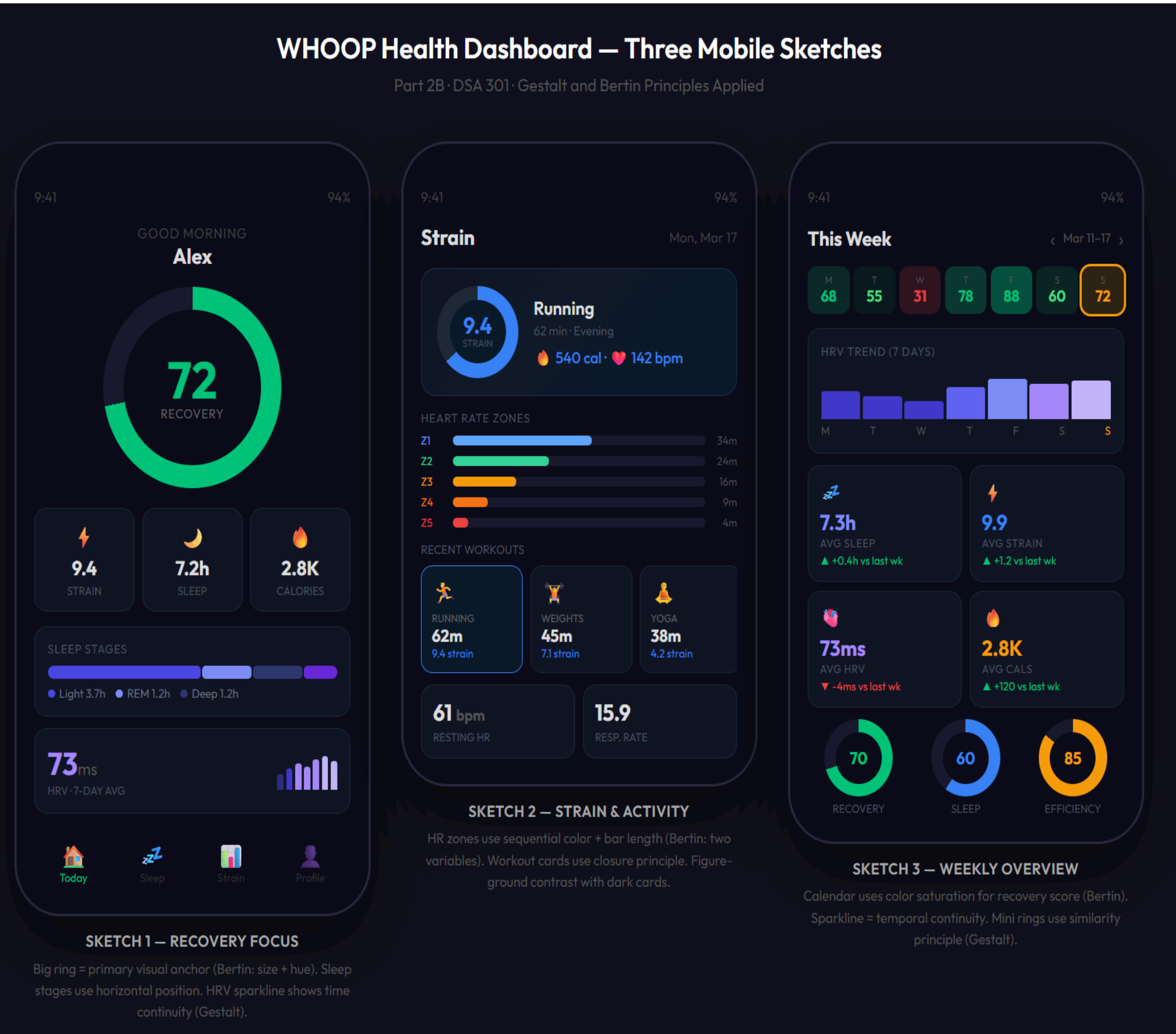
The main element is a big circular ring that shows the recovery score. The ring uses size and hue together (Bertin: redundant encoding). Green means good recovery, amber means okay, red means bad. This helps the user understand with just one look. Below the ring there are three small tiles showing strain, sleep, and calories. These tiles use proximity (Gestalt) to group related daily metrics.

Sketch 2: Strain and Workout

This screen focuses on the workout details. It uses a horizontal bar chart for heart rate zones. The bars use both length and color (Bertin: two retinal variables). Green is low intensity, orange is medium, red is maximum. Below this there is a scrollable list of recent workouts. Each workout is in its own card (Gestalt: closure). The user can scroll horizontally to see more.

Sketch 3: Weekly Overview

This screen shows a full week at once. At the top there is a 7-day calendar strip. Each day is a small square. The background color of each square shows the recovery score using color saturation (Bertin: saturation for quantity). Dark green means very good, light green means okay, red means bad. Below this there is a sparkline chart for HRV trend (Gestalt: continuity). At the bottom there are 4 small metric cards with week over week comparison.



2C | D3.js Implementation

I implemented Sketch 1 using D3.js. I used the same code style as the recitation examples. The implementation has two charts:

Chart 1: Scatter Plot (Recovery vs Sleep Hours)

This chart shows recovery_score on the Y axis and sleep_hours on the X axis. Each circle is one user day record. The color shows the fitness_level of the user (Beginner=red, Intermediate=yellow, Advanced=blue, Elite=green). I used d3.scaleOrdinal() for the color and d3.scaleLinear() for both axes. I also added a tooltip that shows when you hover over a circle.

Chart 2: Bar Chart (Average Recovery by Day)

This bar chart shows the average recovery score for each day of the week. I calculated the average using JavaScript reduce(). The bars use a color gradient from red to green based on the average score (d3.scaleLinear with two colors). This uses Bertin's hue and position variables together.

D3 functions I used:

- d3.csv() — to load the CSV data file
- d3.scaleLinear() — for X and Y axes
- d3.scaleOrdinal() — for fitness level colors
- d3.scaleBand() — for bar chart X axis
- d3.axisBottom() and d3.axisLeft() — for axes
- d3.max() — to find maximum values for scales
- .on("mouseover") — for tooltip interaction

Summary

In this assignment I learned how to use D3.js to make interactive charts from real datasets. The Japan immigration data shows that immigration to Japan grew very fast in the 1980s and 1990s. Most people came from South Korea, Taiwan, and the USA.

The WHOOP health dataset is very useful for a mobile app because it has many different health attributes. The most important ones are recovery score, HRV, and sleep hours. I showed three different ways to visualize this data in a mobile app.

I also learned about Gestalt and Bertin principles. Gestalt principles like proximity and continuity help the user understand the chart naturally. Bertin principles like position, size, and hue help encode data in the most effective way.

Note

All parts of this assignment that does not ask for AI usage are entirely my own work. The written analysis, chart design decisions, D3.js implementation, data processing, and running the local server with Python terminal are all done by me. AI was only used for the grammar checking of the report text. The interactive dashboard creation in Part 1A was done with AI as the assignment ask. For Part 2B, the mobile dashboard sketch ideas and design are my own, but I use AI to help me display it as a HTML file in the browser because the assignment did not ask for this part to be implement in code.